The Thermodynamic Formalism of the Riemann Zeta Function: Part I: Prime Hub Graphs and the Vacuum Trace

Consolidated FST-RH Framework

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Abstract

This paper establishes a correspondence between the Riemann Hypothesis and the statistical mechanics of open dissipative systems. By representing the primes as primitive orbits of a holonomy-saturated directed graph (the "Prime Hub Graph"), we construct a Ruelle transfer operator whose spectral determinant reproduces the Riemann zeta function $\zeta(s)$. The core innovation is the translation of the spectral symmetry required for the Riemann Hypothesis into a requirement of time-reversal symmetry for the underlying Markov dynamics, motivated by the Maximum Entropy Production Principle (MEPP). This first part of a two-part series rigorously details the geometric construction, the holonomy filter, and the derivation of the $1/\zeta(s)$ vacuum trace, while explicitly isolating the physical heuristics and circularities (such as the prime orbit axiom) that remain to be overcome mathematically.

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1 Introduction

The Hilbert-Pólya conjecture proposes that the non-trivial zeros $s = \frac{1}{2} + i\gamma$ of the Riemann zeta function correspond to the eigenvalues of a self-adjoint operator $\hat{H}$ such that $H\psi = \gamma\psi$. The semi-classical Berry-Keating approach $\hat{H} = xp$ fails primarily due to the Weyl law obstruction: the number of zeros of $\zeta(s)$ up to height T grows as $\frac{T}{2\pi} \log \frac{T}{2\pi e}$, while a one-dimensional quantum system yields an unphysical dense point spectrum.

This paper formally shifts the paradigm from quantum mechanics (Hermitian operators, unitary flow) to statistical mechanics (Ruelle transfer operators, dissipative flow). The Riemann zeros are generated not as eigenvalues of a Hamiltonian, but as the zeros of a Selberg-type or Ruelle-type dynamical zeta function.

2 The Prime Hub Graph and Model RH

To reproduce the arithmetic Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$, we require a dynamical system whose primitive periodic orbits map bijectively to the prime numbers.

Definition 2.1 (The Prime Orbit Axiom)**.** We postulate the existence of a topologically mixing directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ such that its primitive cycles γ_p are in one-to-one correspondence with the prime numbers $p \in \mathcal{P}$, with the geometric length of the cycle given by $\ell(\gamma_p) = \log p$. If a cycle γ traverses k times around γ_p, its length is $k \log p$.

Remark 2.2 (Motivation vs. Logical Dependency). The Prime Orbit Axiom serves as a constructive motivation for the thermodynamic formalism developed in this series. It demonstrates that the Riemann zeta function can be interpreted as a vacuum trace of a symbolic dynamical system. However, the analytical results in Part II and Part III regarding Gauge Positivity and the critical identification $\lim m_n = \lambda_n$ depend solely on the properties of $\zeta(s)$ and the arithmetic Euler product, and they remain logically independent of the specific graph construction presented here.

3 The Maximum Entropy Production Principle

The central physical idea is to interpret the critical line $\mathrm{Re}(s) = \frac{1}{2}$ as the condition of detailed balance or time-reversal symmetry. In statistical mechanics, the Maximum Entropy Production Principle (MEPP) posits that non-equilibrium systems tend to configurations that maximize the production of thermodynamic entropy, which corresponds dynamically to states satisfying time-reversal symmetry in their fluctuations (Gallavotti-Cohen fluctuation theorem).

Heuristic Principle 3.1 (Model RH via MEPP). If the prime numbers represent the optimally stable configuration of a dynamical "prime gas" competing for phase space subject only to the constraint of global density (the Prime Number Theorem), the resulting invariant measure (the Gibbs measure) must satisfy time-reversal symmetry. This symmetry forces the zeros of the corresponding dynamical zeta function to lie on the critical axis $\mathrm{Re}(s) = \frac{1}{2}$.

Therefore, proving the Riemann Hypothesis within this framework reduces to proving that the specific transfer operator constructed over the Prime Hub Graph inherently satisfies the symmetry conditions required by MEPP.

4 Holonomy-Saturated Transfer Operators

To decouple the chaotic mixing of the graph from the prime lengths, we introduce a free-monoid holonomy.

Definition 4.1 (Holonomy Filter)**.** Let Γ be the free monoid generated by the primes $\mathcal{P}$. We assign a weight $g_e \in \Gamma$ to each edge $e \in \mathcal{E}$ such that the ordered product of weights around a primitive cycle γ_p is exactly the generator $[p] \in \Gamma$, while non-prime cyclic paths yield non-generator words.

We define the transfer operator $\mathcal{L}_s$ acting on functions $f : \mathcal{V} \times \Gamma \to \mathbb{C}$ equipped with a symbolic state:

$$(\mathcal{L}_s f)(x, g) = \sum_{y \to x} e^{-s\,\ell(x \leftarrow y)} f(y, g \cdot g_{y \to x})$$

4.1 The Vacuum Trace

By taking the trace over the trivial holonomy sector (the "Vacuum Trace" $\mathrm{Tr}_{\mathrm{vac}}$), we project out all chaotic paths that do not form pure prime-power loops.

Theorem 4.2 (Recovery of the Zeta Function)**.** *For $Re(s) > 1$, the Fredholm determinant of the vacuum-projected transfer operator satisfies:*

$$\det(I - \mathcal{L}_s)^{vac} = \prod_p \left(1 - e^{-s \log p}\right) = \prod_p \left(1 - p^{-s}\right) = \frac{1}{\zeta(s)}.$$

Proof. By the Ruelle identity, $\log \det(I - \mathcal{L}_s) = -\sum_{m=1}^{\infty} \frac{1}{m} \mathrm{Tr}(\mathcal{L}_s^m)$. The vacuum trace restricts the sum precisely to configurations where the word formed by the path is the identity in the Abelianization, filtering isolated prime cycles. Thus $\mathrm{Tr}_{\mathrm{vac}}(\mathcal{L}_s^{m \log p}) = \log p \cdot p^{-ms}$ for prime powers, which integrates exactly to $-\log(1 - p^{-s})$. □

5 Conclusion to Part I

Part I has established the mechanical bridge: The zeros of $\zeta(s)$ are the resonances of the vacuum-projected transfer operator over the Prime Hub Graph. The Riemann Hypothesis asserts that these resonances are symmetrically distributed around $\mathrm{Re}(s) = \frac{1}{2}$. While the MEPP heuristic explains *why* nature would select such a symmetric configuration, it does not constitute a mathematical proof.

In Part II, we discard the heuristics and attack the mathematical obstructions directly: Li's Criterion, excess free energy, the convexity trap, and the reduction of the RH to the gauge positivity of an arithmetic kernel.